

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		Labels on diagram to show vertical condenser (unsealed at top) (1) Water in and out of condenser in correct direction (1) Heat source below flask with reagents (1)	3			3		3
		(ii)		Liquid evaporates, vapour goes into condenser, cools and goes back to liquid / condenses	1			1		1
		(iii)		Any of following for (1) - The reaction is slow - Allows time for equilibrium to be established - Stops reactants / products boiling away			1	1		1
		(iv)		Catalyst/ dehydrating agent	1			1		
	(b)	(i)		Fractional distillation (1)	1			1		1
		(ii)		Moles of ethanoic acid = 0.05 and moles of methanol = 0.04 (1) Theoretical yield of methyl ethanoate = $0.04 \times 74 = 2.96 \text{ g}$ (1) % of theoretical yield = $\frac{1.18}{2.96} \times 100 = 40 \%$ (1) ecf possible		3		3	3	

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		(iii)		Reflux for longer (1) Reaction is slow/ needs time to establish equilibrium (1) or Add extra methanol / ethanoic acid (1) To allow more of the acid to react/ push the equilibrium to RHS (1)			2	2		2
	(c)	(i)		Dehydration/ elimination	1			1		
		(ii)		Displayed formulae of butan-1-ol and butan-2-ol for (1) each H H H H H-C-C-C-C-H H H H OH butan-1-ol H H H H H-C-C-C-C-H H H OH H butan-2-ol		2		2		
		(iii)		Orange to green	1			1		
		(iv)		Oxidation of alcohol / redox	1			1		

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		(v)		Displayed formula of butanoic acid / butanal / butanone H H H O H-C-C-C-C H H H O-H H H H O H-C-C-C-C H H H H H H O H H-C-C-C-C H H H H		1		1		
				Question 11 total	9	6	3	18	3	8